CMP6200/DIG6200

Individual Undergraduate Project (FYP)

Interim Progress Review

Augmented Reality(AR) based Virtual Tour Guide

Student Name: Sadikshya Ghimire

Student ID: 23140745

Course: CMP6200 Individual Honours Project

Supervisor: Sujan Shrestha

Reporting Period: February 1,2026 to April 24,2026

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1. Log of Completed Activities & Milestones

This report is the compilation of all the technical and non technical work that I have done during the period of the review of February 1,2026 to April 24,2026

Aspect	Description
Current Project Phase	Advanced development and early integration phase, transitioning from experimentation to a working prototype
UI Development	Core user interface fully completed and functional
Chatbot (NLP Integration)	Functional chatbot developed using API-based NLP, enabling interactive user queries
Computer Vision Model	Landmark recognition model successfully trained using prepared dataset
AR Development (Unity)	Partially implemented; currently testing camera feed and real-time AR overlay behavior
System Integration Status	In progress; focus on connecting CV, AR, and chatbot into a unified system
Development Approach	Shift from isolated module development to integrated system architecture
Key Achievement	Successful development of all core components required for the system
Current Focus	Ensuring seamless interaction between modules and system-level functionality
Significant Insight	Identified and addressing the challenge of integrating multiple AI technologies into one cohesive application
Alignment with Proposal	Directly addresses the problem of fragmented AI systems in tourism applications

Figure 1: Project Progress Overview

2.Artefact Design and Development

The aim of this section is to demonstrate progress made till date, and to summarise project challenges and their solutions through the project lifecycle.

2.1 Progress Against Original Timeline

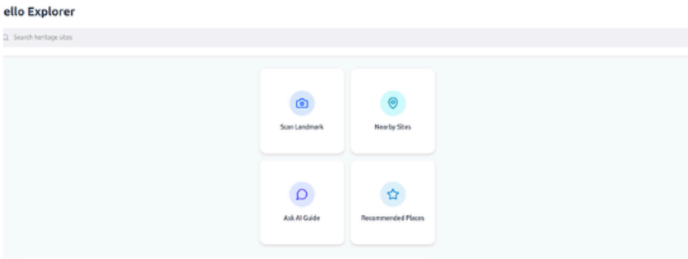
Component	**Status**	**Description**
Dataset Collection	Completed	3500–4000 images across all heritage sites
Image Preprocessing	Completed	Augmentation, resizing, normalization
Model Training (CV)	Completed	Lightweight model trained for mobile deployment
UI Design	Completed	Fully functional mobile interface
Chatbot (NLP API)	Completed	Context-aware chatbot integrated
AR Environment (Unity)	In Progress	Camera + AR overlay under testing
System Integration	In Progress	Components developed separately, integration pending
Testing & Validation	In Progress	Initial testing done, full testing pending

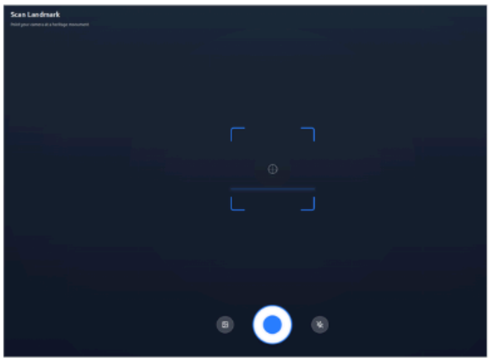
Figure 2: Progress Against Original Timeline

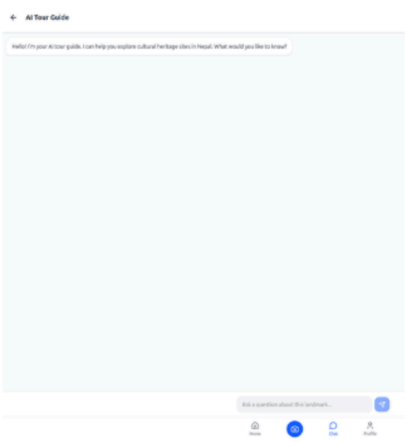
The project is 80% complete due to the Gantt chart and the project completed milestones:

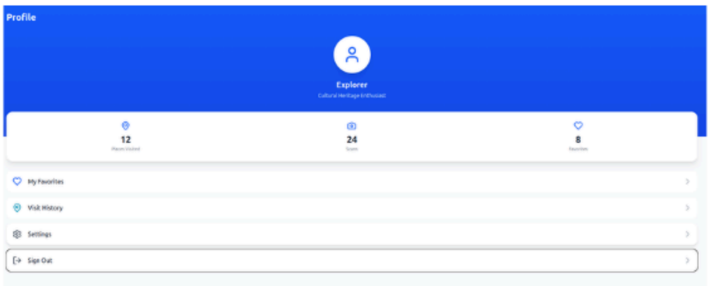
☑ On Track

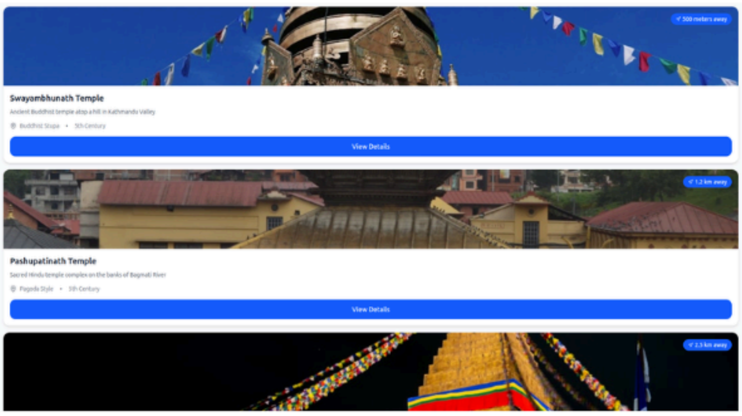
2.2 Implementation and Current State



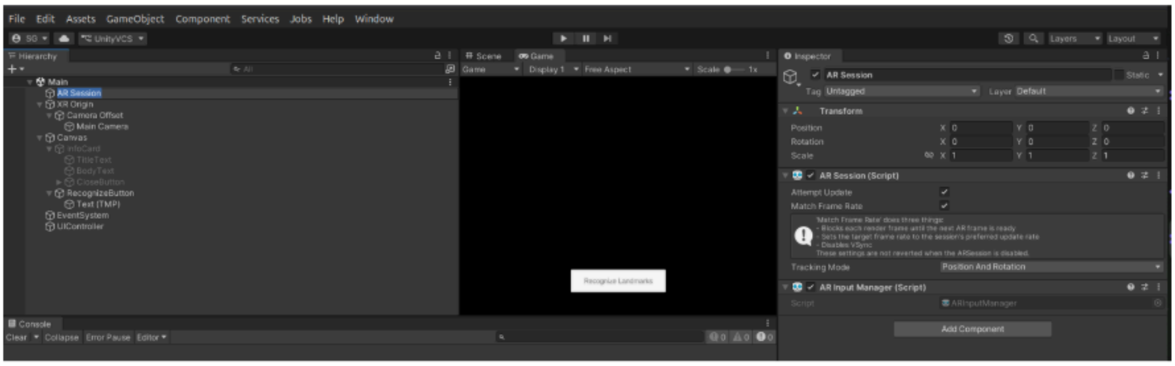


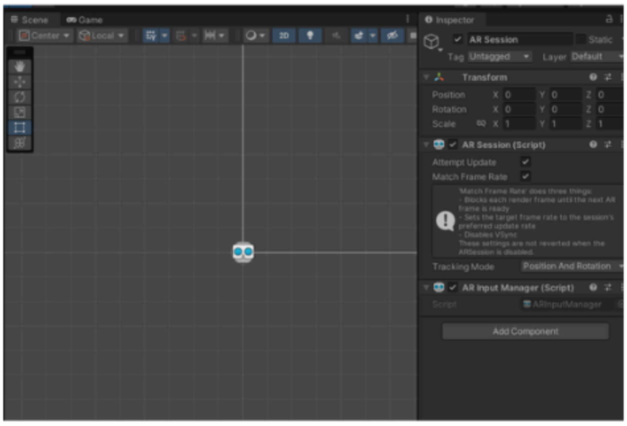






Chatbot UI





Unity Interface and Scene play

2.4 Challenges Encountered & Resolutions

I encountered some challenges, which i have mentioned below and what actions i took in order to solve them

Challenge	Impact	Resolution
Fragmented development	Delayed integration	Shifted to integration-focused approach
Unity AR complexity	Slower development	Incremental testing in Unity scenes
Multi-tech stack (CV + AR + NLP)	High system complexity	Modular debugging strategy
Time pressure	Risk to completion	Prioritizing core features only

Figure 4: Challenges Encountered & Resolutions

3. Action Plan: Goals for Next Period

The full system integration will be completed by connecting the computer vision model, Unity-based AR module, and chatbot into a single functional application. The Unity AR environment will be finalised to ensure stable camera tracking and accurate real-time information overlays. Comprehensive system testing will be conducted, including functional testing and validation of real-time performance, to verify that all components operate reliably together. The landmark recognition model will be further improved and evaluated using unseen test data to assess its accuracy and robustness. In addition, a basic content-based recommendation system will be implemented in line with the project objectives. Report writing will be completed with a focus on the implementation, results, and evaluation chapters, while a fully working prototype and demonstration will be prepared for the final presentation and viva.

4. Questions for Supervisor

1) This is composed of computer vision, AR, and a chatbot, what would be the metrics that you would suggest for evaluating the whole system?
2) Since the accuracy of about 80% in landmark recognition appears to be enough for this project, what should be a greater priority, improving robustness under challenging realistic scenarios?

5. Conclusion and Reflection

This project has gone from the conceptualization and planning phase into the main elements of development, in which many of the elements are already designed. The current priority is to update all the components – an essential component where the planned features are combined into a functional application.

The project has also emphasized the importance of not just developing the technologies, but also integrating them effectively to provide a seamless and enjoyable tourist experience. This supports the project's aim of bringing together different AI technologies into a single solution for tourism.

Overall, the project is progressing well, with clear objectives and priorities in place. As long as the integration and testing phase are successful it promises for a successful completion within the planned time.